Balanced transport at the egalitarian core: a complete boundary phase diagram

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Abstract

For a transferable-utility game, exact partition at an allocation means that every cumulative union of its richest payoff strata is tight. Both this condition and its Lorenz consequence are due to earlier work. We determine exactly when the condition can fail at the leximin core allocation of a singleton-normalized exact game. If a strict payoff boundary has m cumulative poorer players and h richer players, then a slack richer suffix exists if and only if

$$m \geq m_{\min}(h), \qquad m_{\min}(1) = 4, \quad m_{\min}(2) = m_{\min}(3) = 3, \quad m_{\min}(h) = 2 \ (h \geq 4).$$

The lower bounds are structural. A normal-cone theorem turns leximin optimality into one simultaneous strict cover by tight coalitions. Projecting the cover to the richer side and applying a convex defect functional shows more generally that every failure selects one lifted minimal balanced packet with a positive rooted semi-balanced obstruction; mixtures are unnecessary. The resulting certificate uses at most $|N|-1$ positive coalitions, and its slack budget forces some one-player extension of the richer suffix to be slack as well. One poorer player cannot support the required opposing signs. A unique richest player needs four poorer players by a fractional sacrifice budget. With two poorer players, every failure needs a proper active hyperedge of size at least three, hence at least four richer players. Three explicit dimension-free lower-envelope families attain every remaining pair. The familiar four-player guarantee and sharp five-player failure are immediate corollaries. The argument enumerates no games and no payoff-stratum patterns; its only case analyses are over the shapes of a protector trace and over the components of a rank-two vertex.

1 Introduction and prior-art boundary

Let (N, v) be a transferable-utility game and

$$\mathrm{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for every } S \subseteq N\}.$$

The game is *exact* if every coalition inequality is attained by some core allocation. Write $\phi = \mathrm{Lmin}(v)$ for the unique core allocation whose increasing rearrangement is lexicographically maximal. At a strict payoff boundary of ϕ, let L be the cumulative poorer side and let $H = N \setminus L$ be the richer suffix.

The exact-partition condition and its implications for Dutta–Ray and Lorenz selection are prior art (Llerena and Mauri, 2017; Sánchez-Soriano et al., 2010). Qualitative tight-coalition descriptions of the egalitarian selections are likewise established: Arin and Iñarra (2001) characterize the egalitarian core by the impossibility of a rich-to-poor transfer inside the core, which is exactly the existence of a tight coalition containing the richer player and omitting the poorer one, and they place Lmin in that set; Arin et al. (2003) characterize Lmax on large-core games; see also

Mauri (2018). Exact balancedness and semi-balanced facet systems are also established (Csóka et al., 2011; Lohmann et al., 2012; Studený and Kratochvíl, 2022). In particular, the latter authors already obtain purely minimal semi-balanced systems from minimal balanced systems by complementing the exceptional coalition. Our candidate contribution is not that full-set correspondence, but the Lmin-selected normal cover and its projection to one minimal balanced packet on the richer suffix, together with the sharp two-parameter classification and realizing families. The closest published low-dimensional result known to us is that exact games with at most four players have large cores (Biswas et al., 1999) and that large-core games are egalitarian stable, with PES = Lmax on that domain (Dietzenbacher, 2021); that chain does not itself assert exact partition at Lmin.

The main theorem is stated now. All games below are singleton-normalized, meaning $v(\{i\}) = 0$ for every player.

Theorem 1 (boundary phase diagram)**.** *At a strict payoff boundary of $\phi = \mathrm{Lmin}(v)$ in a singleton-normalized exact game, put $m = |L|$ and $h = |H|$. It is possible that $v(H) < \phi(H)$ exactly for*

$$m \geq m_{\min}(h), \qquad m_{\min}(h) = \begin{cases} 4, & h = 1, \\ 3, & h = 2 \text{ or } 3, \\ 2, & h \geq 4. \end{cases} \tag{1}$$

Every admitted pair is realized by an integral indecomposable exact game. The poorer side may contain several payoff strata.

The proof separates a general lower-bound mechanism from three universal constructions. This makes the four/five threshold a corollary rather than the theorem's organizing principle.

2 Strict covers and balanced-trace transport

For $\phi \in \mathrm{Core}(v)$ write

$$s_\phi(S) = \phi(S) - v(S).$$

A coalition is ϕ-tight when its slack is zero.

Lemma 2 (strict richer cover)**.** *Let $\phi = \mathrm{Lmin}(v)$ and let $N = L \,\dot\cup\, H$ be a strict payoff boundary. There are nonnegative weights on proper ϕ-tight coalitions whose incidence load is a common positive number on H and is strictly smaller at every player in L.*

Proof. Let $\mathcal{D}$ be the tangent cone of the core at ϕ. There is no $d \in \mathcal{D}$ with $d_i \geq 0$ for every $i \in L$ and one of these inequalities strict: a sufficiently short segment $\phi + td$ remains in the core, preserves the strict gap between L and H, and improves the first poorer payoff stratum on which d is nonzero.

Let $K \subseteq \mathbb{R}^L$ be the image of $\mathcal{D}$ under the coordinate projection. It is a polyhedral cone, being the linear image of a polyhedral cone, and the previous paragraph says $K \cap \mathbb{R}^L_+ = \{0\}$.

We claim the polar cone $K^\circ = \{w \in \mathbb{R}^L : \langle w, y\rangle \leq 0 \text{ for all } y \in K\}$ contains a *strictly* positive vector. Ordinary separation does not give this: two cones meeting only at the origin admit a separating $w \neq 0$ with $w \geq 0$, and such a w may vanish on some coordinates. Take polars instead. Polarity turns the intersection into a sum, so

$$\mathbb{R}^L = \{0\}^\circ = (K \cap \mathbb{R}^L_+)^\circ = \mathrm{cl}\big(K^\circ + (\mathbb{R}^L_+)^\circ\big) = \mathrm{cl}\big(K^\circ + \mathbb{R}^L_-\big).$$

Both K° and $\mathbb{R}^L_-$ are polyhedral, and a sum of finitely many polyhedral cones is polyhedral, hence closed; the closure operation is therefore vacuous and $K^\circ + \mathbb{R}^L_- = \mathbb{R}^L$. Write $\mathbf{1}_L = w + p$ with

$w \in K^\circ$ and $p \le 0$. Then $w = \mathbf{1}_L - p \ge \mathbf{1}_L > 0$. Thus $w_i > 0$ for every $i \in L$ and $\sum_{i\in L} w_i d_i \le 0$ for all $d \in \mathcal{D}$.

Extend w by zero on H; the displayed inequality is unchanged, so w lies in the polar of $\mathcal{D}$ in $\mathbb{R}^N$. Since $\mathcal{D} = \{d : d(N) = 0,\ d(T) \ge 0 \text{ for every proper } \phi\text{-tight } T\}$, that polar is generated by $\pm\mathbf{1}_N$ and by $-\mathbf{1}_T$, giving

$$w = \mu\mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T, \qquad \lambda_T \ge 0, \tag{NC}$$

where T ranges over proper ϕ-tight coalitions. Thus the weighted incidence load is μ on H and $\mu - w_i < \mu$ at $i \in L$. Nonnegativity of the load and positivity of w_L imply $\mu > 0$. Deleting coalitions disjoint from H preserves the richer load and only lowers poorer loads. □

Normalize the common rich load in Lemma 2 to one and remove coalitions disjoint from H. Project every remaining tight coalition to its nonempty trace on H. The projected weights lie in the exact-cover polytope

$$\mathcal{P}_H = \left\{ \beta \ge 0 : \sum_{\emptyset \ne R \subseteq H} \beta_R \mathbf{1}_R = \mathbf{1}_H \right\}. \tag{2}$$

Its vertices are the unique normalized weights of minimal balanced collections on H; this is standard balanced-collection geometry (Lohmann et al., 2012; Laplace Mermoud et al., 2025).

Theorem 3 (balanced-trace transport)**.** *Suppose $s = s_\phi(H) > 0$. Call a* lifted packet *a pair consisting of a vertex β of $\mathcal{P}_H$ together with a choice, for each trace R in its support, of one supported ϕ-tight coalition whose trace on H is R; its* poor-load vector *$r \in \mathbb{R}_+^L$ records the total weight of chosen coalitions containing each $i \in L$. Then there are a core point z, a displacement $\Delta = z - \phi$, and a finite convex mixture $\{(\theta_q, r^q)\}$ of lifted packets such that*

$$\Delta(H) = -s, \qquad \Delta(L) = s, \tag{3a}$$

$$\min_{i\in L} \Delta_i < 0 < \max_{i\in L} \Delta_i, \tag{3b}$$

$$-s + \sum_{i\in L} r_i^q \Delta_i \ge 0 \qquad \text{for every } q, \tag{3c}$$

$$\sum_q \theta_q r_i^q < 1 \qquad \text{for every } i \in L. \tag{3d}$$

Proof. Exactness supplies $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$, proving (3a). Let $c_i < 1$ be the poorer loads of a normalized cover from Lemma 2. Every supported coalition is tight at ϕ and feasible at z, hence

$$0 \le \sum_T \lambda_T \Delta(T) = -s + \sum_{i\in L} c_i \Delta_i. \tag{4}$$

If every poorer displacement were nonnegative, the right side would be strictly smaller than $-s + \Delta(L) = 0$. Since $\Delta(L) = s > 0$, both signs occur, proving (3b).

Let $\beta_R = \sum_{T\cap H = R} \lambda_T$ be the projected weights, so $\beta \in \mathcal{P}_H$, and decompose $\beta = \sum_p \theta_p \beta^p$ into vertices. Independently for each trace R, choose a preimage T with $T \cap H = R$ with conditional probability λ_T/β_R. A vertex together with such a section is a lifted packet, and the product measure assigns it the weight $\theta_p \prod_R (\lambda_{T_R}/\beta_R)$; index the resulting finite mixture by q.

Fix one lifted packet q, with vertex β^p and section σ. Each $\sigma(R)$ is tight at ϕ and feasible at z, so $\sum_R \beta_R^p \Delta(\sigma(R)) \ge 0$. A rich player j lies in $\sigma(R)$ exactly when $j \in R$, so its load is $\sum_{R\ni j} \beta_R^p = 1$, while a poor player i has load r_i^q by definition. The displayed sum therefore equals $\Delta(H) + \sum_{i\in L} r_i^q \Delta_i = -s + \sum_{i\in L} r_i^q \Delta_i$, proving (3c) for every packet, not merely on average.

For (3d), average r_i over the product measure. Since $\sum_p \theta_p \beta_R^p = \beta_R$ and the section is chosen independently per trace,

$$\sum_q \theta_q r_i^q = \sum_R \beta_R \cdot \frac{1}{\beta_R} \sum_{T \cap H = R,\ i \in T} \lambda_T = \sum_{T \ni i} \lambda_T = c_i < 1.$$

□

The mixture in Theorem 3 always contains one packet that already carries a positive rooted obstruction. This gives a general dimension-free reduction before the cardinality arguments begin.

Theorem 4 (one-packet reduction)**.** *At every slack strict boundary there is a lifted vertex of $\mathcal{P}_H$, with poorer load vector $r \geq 0$, such that, on putting*

$$b = \max_{i \in L} r_i, \qquad u_i = b - r_i, \qquad \delta = 1 + \sum_{i \in L} u_i - b,$$

one has $\delta > 0$ and

$$\sum_T \lambda_T \mathbf{1}_T + \sum_{i \in L} u_i \mathbf{1}_{H \cup \{i\}} - \delta \mathbf{1}_H = b \mathbf{1}_N. \tag{P}$$

Here the T are proper ϕ-tight coalitions and their traces form one minimal balanced collection on H. Moreover,

$$\delta s_\phi(H) \leq \sum_{i \in L} u_i s_\phi(H \cup \{i\}), \tag{PB}$$

so at least one u_i is positive. The trace collection is either the full-trace packet $\{H\}$ or consists entirely of proper subsets of H. The identity uses at most $|H|$ packet coalitions and at most $|L| - 1$ spokes, hence at most $|N| - 1$ positive coalitions.

Proof. Let $c_i = \sum_q \theta_q r_i^q < 1$ be the average loads in (3d). For a nonnegative vector a define

$$F(a) = 1 + (|L| - 1) \max_i a_i - \sum_i a_i.$$

This function is convex, and $F(c) \geq 1 - \max_i c_i > 0$. Jensen's inequality therefore gives a lifted vertex with $F(r) > 0$; this is exactly the displayed δ. Its rich load is one. In (P), poor player i has load $r_i + u_i = b$, while every rich player has load $1 + \sum_i u_i - \delta = b$.

The positive incidence columns at a vertex of $\mathcal{P}_H$ are linearly independent: otherwise a sufficiently small signed kernel perturbation would remain nonnegative in both directions. Thus the packet has at most $|H|$ traces. The pure lift uses one coalition per trace. Any i attaining the maximum b has $u_i = 0$, so at most $|L| - 1$ spokes are positive, proving the sparsity bound.

Choose $z \in \mathrm{Core}(v)$ exact on H. Evaluate (P) at z and at ϕ. Writing $\Delta = z - \phi$, the packet terms satisfy $\Delta(T) \geq 0$, the spokes satisfy $\Delta(H \cup \{i\}) \geq -s_\phi(H \cup \{i\})$, and $\Delta(H) = -s_\phi(H)$. The displacement form of (P) is therefore

$$0 = \sum_T \lambda_T \Delta(T) + \sum_i u_i \Delta(H \cup \{i\}) - \delta \Delta(H) \geq -\sum_i u_i s_\phi(H \cup \{i\}) + \delta s_\phi(H),$$

which is (PB). If every u_i were zero, (PB) would contradict $\delta s_\phi(H) > 0$. Finally, a minimal balanced collection containing H must be the singleton collection $\{H\}$; otherwise all its traces are proper. □

Thus every failure has an exterior full-trace packet or an internal minimal balanced-transport packet. The incidence identity is an established semi-balanced system with exceptional coalition H; the new point proposed here is that the Lmin normal certificate and the convex functional F select one minimal balanced packet on H from an arbitrary failure. This is different from the established complement-of-the-exceptional-set correspondence, which operates on the full player set.

Remark 5 (locating the packets in the published catalogue)**.** *Every system (P) is semi-balanced with exceptional coalition H in the sense of Studený and Kratochvíl (2022), and is minimal in every instance computed for this paper. Their electronic catalogue lists the permutational types of the* indecomposable *min-semi-balanced systems on at most six players, sixteen on five players and fifty-three on six; these are exactly the types that index facets of the cone of exact games. Indecomposability is here a property of the set system and is unrelated to direct-sum indecomposability of a game. Five of the five-player types and fifteen of the six-player types have a singleton exceptional coalition, which is necessary for an $h = 1$ obstruction. It is not sufficient: at $h = 1$ the exact-cover polytope (2) has the single vertex $\{H\}$ with weight one, so a system (P) has one lift, at most one positive coalition of size exceeding two, and all positive coefficients equal to one. Only three of the five singleton-exceptional five-player types have that form; the other two are never systems (P), at any boundary of any exact game on any number of players.*

Take the smallest realization of each of the three minimal pairs of the staircase (1). For the $(4, 1)$ member of Proposition 14 with richest player e, the tight proper coalitions at ϕ are ce, de, cde, $acde$, $bcde$, and every admissible packet coalition produces a catalogued type:

$$\begin{aligned} T = ce \text{ or } de: &\quad \mathbf{1}_{ae} + \mathbf{1}_{be} + \mathbf{1}_{ce} + \mathbf{1}_{de} - 3 \cdot \mathbf{1}_{e} = \mathbf{1}_N, \\ T = cde: &\quad \mathbf{1}_{ae} + \mathbf{1}_{be} + \mathbf{1}_{cde} - 2 \cdot \mathbf{1}_{e} = \mathbf{1}_N, \\ T = acde: &\quad \mathbf{1}_{acde} + \mathbf{1}_{be} - \mathbf{1}_{e} = \mathbf{1}_N. \end{aligned}$$

For the $(3, 2)$ member of Proposition 16, where $\phi = (1, 1, 2, 3, 3)$ and $H = de$, the only tight proper coalitions are ade and cde; both lifts give the full-trace packet, and the single resulting type is again catalogued:

$$\mathbf{1}_{ade} + \mathbf{1}_{bde} + \mathbf{1}_{cde} - 2 \cdot \mathbf{1}_{de} = \mathbf{1}_N.$$

For the $(2, 4)$ member of Proposition 15, where $\phi = (2, 2, 4, 4, 4, 4)$ and $H = cdef$, three types occur:

$$\begin{aligned} &\textit{catalogued:} && \mathbf{1}_{acdef} + \mathbf{1}_{bcdef} - \mathbf{1}_{cdef} = \mathbf{1}_N, \\ &\textit{not catalogued:} && \mathbf{1}_{ac} + \mathbf{1}_{def} + \mathbf{1}_{bcdef} - \mathbf{1}_{cdef} = \mathbf{1}_N, \\ &\textit{not catalogued:} && \mathbf{1}_{cef} + \mathbf{1}_{def} + \mathbf{1}_{acde} + \mathbf{1}_{acdf} + 2 \cdot \mathbf{1}_{bcdef} - 3 \cdot \mathbf{1}_{cdef} = 2 \cdot \mathbf{1}_N. \end{aligned}$$

The last two systems are min-semi-balanced but decomposable, by $E = \{c\}$ and by $E = \{c, d, e\}$; for the second system, for instance, $\mathbf{1}_{ac} + \mathbf{1}_{bcdef} - \mathbf{1}_{c} = \mathbf{1}_N$ is a combination with nonnegative coefficients inside the system and a negative one on E. Two points follow. First, the certificate of Theorem 4 is always a valid inequality for the exact cone but need not be a facet, so being catalogued is strictly stronger than being a system (P). In the instances computed, non-facet certificates arose only in the two-poorer family. Second, a single failure does not select a single class: which class appears depends on the tight coalition the cover supports. This is the concrete form of the multi-class phenomenon noted in Section 6. Decomposability is a statement about set systems alone; here $bcdef$ carries positive slack at ϕ, so it exhibits no sharper certificate at this boundary.

Two consequences are worth isolating. The first records what the certificate costs; the second is a propagation statement that needs no cardinality hypothesis.

Corollary 6 (linear-size witness and slack propagation)**.** *Let a strict payoff boundary of $\phi = \mathrm{Lmin}(v)$ have $s_\phi(H) > 0$. Then:*

(i) *the obstruction is exhibited by at most* $|N|-1$ *positive coalitions together with* H*, and by the rational data* λ, u, δ, b *of (P);*

(ii) $s_\phi(H) \le \dfrac{1}{\delta}\displaystyle\sum_{i\in L} u_i s_\phi(H\cup\{i\})$*;*

(iii) *consequently* $s_\phi(H\cup\{i\}) > 0$ *for at least one* $i \in L$*.*

Proof. Part (i) is the sparsity bound of Theorem 4. Part (ii) is (PB) divided by $\delta > 0$. For (iii), the right side of (ii) is positive because the left side is, and every u_i is nonnegative, so some product $u_i s_\phi(H\cup\{i\})$ is positive. □

Part (iii) says that a slack richer suffix is never isolated: the failure propagates to a one-player enlargement of H. Since the enlargement is still a proper coalition, iterating is not possible in general, and the statement is genuinely about the single step. Part (ii) quantifies this: the slack at H is financed, at exchange rate δ, by the slacks of at most $|L|-1$ single-player completions.

The first exclusion is immediate.

Corollary 7. *A slack boundary cannot have* $m = 1$*.*

Proof. Equation (3b) requires both a positive and a negative displacement on L. □

3 The two remaining lower bounds

3.1 A unique richest player

Assume $H = \{r\}$, write $\phi_r = \gamma$, and put $q_i = \phi_i < \gamma$ for $i \in L$. The transfer-blocking characterization of the egalitarian core (Arin and Iñarra, 2001) gives, for each $i \in L$, a tight protector

$$T_i = \{r\}\cup Q_i, \qquad \emptyset \ne Q_i \subseteq L\setminus\{i\}. \tag{5}$$

The nonemptiness follows because the normalized singleton $\{r\}$ is slack. Alternatively (5) is the case $H = \{r\}$ of Lemma 2: the cover has load μ at r and load $\mu - w_i < \mu$ at i, so the supported coalitions containing r but not i carry total weight $w_i > 0$ and at least one of them is positive. The argument below therefore does not depend on the cited characterization.

Lemma 8 (sacrifice budget)**.** *For weights* $\lambda_i \ge 0$ *satisfying*

$$c_j := \sum_{i: j\in Q_i} \lambda_i \le 1, \qquad \Lambda := \sum_i \lambda_i,$$

one has

$$\gamma(\Lambda - 1) \le \sum_{j\in L}(1-c_j)q_j. \tag{6}$$

Proof. Exactify $\{r\}$ at a core point z and put $d = z - \phi$. Summing the tight protector inequalities with the displayed weights, using efficiency and the singleton lower bounds $z_j \ge 0$, gives (6). Equivalently, complete the uncovered poorer loads by singleton coalitions; together with negative weight $\Lambda - 1$ on $\{r\}$, the incidence vector is exact-balanced, and the exact-balanced slack inequality is exactly (6). □

Proposition 9. *If* $h = 1$ *and the suffix is slack, then* $m \ge 4$*.*

Proof. For $m = 1$ no protector in (5) exists. For $m = 2$, both protector traces are the opposite singleton; weight both by one in (6), obtaining $\gamma \le 0$.

Let $m = 3$. If some $Q_i = \{j\}$, weight the protectors indexed by i and j by one. Their traces are disjoint and leave at most one player u uncovered, so (6) gives either $\gamma \le 0$ or $\gamma \le q_u < \gamma$. If no Q_i is a singleton, then $Q_i = L \setminus \{i\}$ for all i. Weight all three protectors by $1/2$; every poorer player has load one and (6) gives $\gamma/2 \le 0$. □

3.2 Exactly two poorer players

We first isolate the exact-face inequality used in the rounding argument.

Lemma 10 (exact-face overload)**.** *Let v be singleton-normalized and exact, let $\phi \in \mathrm{Core}(v)$, and let $D \subseteq H \subsetneq N$. Suppose nonnegative weights on proper ϕ-tight coalitions have load $1 + q$ on D, load one on $H \setminus D$, and loads $0 \le c_i \le 1$ on $i \in N \setminus H$, where $q > 0$. Then*

$$q s_\phi(D) \le \sum_{i \notin H} (1 - c_i)\phi_i. \tag{7}$$

Proof. Choose $x \in \mathrm{Core}(v)$ with $x(D) = v(D)$ and put $d = x - \phi$. Summing the supported tight-coalition inequalities at x gives

$$0 \le d(H) + q d(D) + \sum_{i \notin H} c_i d_i.$$

Efficiency replaces $d(H)$ by $-\sum_{i \notin H} d_i$. Hence

$$q s_\phi(D) = -q d(D) \le \sum_{i \notin H} (1 - c_i)(-d_i) \le \sum_{i \notin H} (1 - c_i)\phi_i,$$

where the last inequality uses $x_i \ge v(\{i\}) = 0$. □

Put $L = \{a, b\}$ and label the displacement in Theorem 3 so that $\Delta_a < 0 < \Delta_b$. By (3d), some lifted extreme packet has $r_b < 1$.

Lemma 11 (proper-hyperedge obstruction)**.** *Every such underloaded packet contains a proper rich trace R with*

$$3 \le |R| \le |H| - 1. \tag{8}$$

Proof. First dispose of the full-trace packet, for which the assertion would be vacuous. If H is itself a supported trace then $\{H\}$ is already balanced on H, so minimality forces the packet to be $\{H\}$ and there is no proper trace at all. This cannot happen at an underloaded packet. Indeed the single lifted coalition T is proper with $H \subseteq T$, hence $T \in \{H,\ H \cup \{a\},\ H \cup \{b\}\}$ and the poor load is $r \in \{(0,0), (1,0), (0,1)\}$. Underloading gives $r_b < 1$, so $r_b = 0$, and (3c) becomes $-s + r_a \Delta_a \ge 0$. That fails for $r_a = 0$ because $s > 0$, and fails for $r_a = 1$ because $\Delta_a < 0$. Every supported trace of the selected packet is therefore a proper subset of H.

Now suppose that every such trace has size at most two. The positive incidence columns of a vertex of (2) are independent. Its rank-two components are therefore isolated singletons of weight one, isolated edges of weight one, and vertex-disjoint odd cycles whose edges have weight $1/2$. This is the classical description of the vertices of a fractional perfect matching polytope, and follows by leaf deletion. In a component, independence forces at most as many positive traces as vertices. A degree-one vertex j lies in a unique trace R, and the balance equation at j forces $\beta_R = 1$; the

balance equations at the other endpoint of R then kill every remaining trace there, so R splits off as an isolated edge, or as an isolated singleton when $R = \{j\}$. Iterating removes all leaves, so a positive singleton cannot persist inside a larger component and every surviving component is a single cycle. An even cycle has dependent incidence columns, hence is excluded; an odd cycle has nonsingular incidence matrix whose unique balanced solution is the uniform weight $1/2$.

An integral packet has integer b-load, and (3c) forces that load to be at least one. Hence the selected packet is fractional. Its poor loads are half-integers. From

$$-s + r_a\Delta_a + r_b\Delta_b \geq 0, \qquad \Delta_a + \Delta_b = s,$$

and $r_b < 1$, one obtains $r_b = 1/2$ and $r_a = 0$. Exactly one half-weight edge on one odd cycle therefore carries b, and no supported coalition carries a.

Round that labeled odd cycle alternately, starting with the labeled edge, and leave every other component at its original weight. The resulting nonnegative combination of proper ϕ-tight coalitions has rich load $\mathbf{1}_H + e_d$ for one rich player d, poor load e_b, and no load on a. Exactify singleton d. Summing the selected tight inequalities and adding the omitted singleton a gives the exact-face overload inequality $\phi_d \leq \phi_a$, contradicting the strict boundary. Indeed, Lemma 10 has $D = \{d\}$, $c_b = 1$, and $c_a = 0$. Thus (8) holds. □

Proposition 12. *If $m = 2$ and the suffix is slack, then $h \geq 4$.*

Proof. Lemma 11 requires a proper trace of size at least three. □

The proof so far yields every forbidden pair in (1) without enumerating games or payoff-stratum patterns.

4 Three universal realizing families

We use a standard lower-envelope observation.

Lemma 13 (lower-envelope realization)**.** *Let G be a finite family of nonnegative allocations with common total T, and define $v(S) = \min_{g\in G} g(S)$. Then every $g \in G$ belongs to* Core(v)*, every coalition attains its worth, and v is exact. If every coordinate is zero in some row, then v is singleton-normalized.*

Proof. The defining inequality gives $g(S) \geq v(S)$ for every g, S, so every row is a core allocation. Conversely every coalition minimum is attained at a row. The last assertion is immediate. □

We record the families compactly. The displayed face inequalities give short progressive leximin proofs; no search is involved.

Proposition 14 (unique-richest family)**.** *For every $m \geq 4$ there is an integral indecomposable exact game with m poorer players $P = \{1, \ldots, m\}$, one richer player r, and*

$$\mathrm{Lmin}(v) = \mathbf{1}_P + (m-2)e_r, \qquad v(\{r\}) = 0.$$

Proof. Apply Lemma 13 to the four common-total rows

$$\begin{aligned} g^1 &= (m-1)(e_4 + e_r),\\ g^2 &= (m-1)(e_3 + e_4),\\ g^3 &= e_1 + e_2 + e_3 + (m-4)e_5 + (m-1)e_r,\\ g^4 &= \mathbf{1}_P + (m-2)e_r, \end{aligned}$$

where the zero e_5 term is omitted at $m = 4$. Every coordinate vanishes in a row. The four attained faces

$$v(N \setminus \{1\}) = v(N \setminus \{2\}) = 2m - 3, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1$$

show that any core point giving everyone at least one fixes players $1, 2$ at one; subtracting $\mathbf{1}_N$ and comparing the two pair inequalities with efficiency then fixes every poorer player at one. The remaining total gives $x_r = m - 2$. Hence $g^4 = \mathrm{Lmin}(v)$. Row g^2 proves $v(\{r\}) = 0$. For indecomposability, observe that

$$v(\{1, 4\}) = v(\{2, 4\}) = v(\{3, 4\}) = 1, \qquad v(\{3, r\}) = v(\{4, r\}) = m - 1,$$

and, when $m \geq 5$, $v(\{4, 5\}) > 0$. These positive pair worths place players $1, \ldots, 5, r$ in one direct-sum component. If $m = 6$, the coalition $T = \{3, 4, 5, r\}$ satisfies $v(T) = 7 < v(T \cup \{6\}) = 8$. If $m \geq 7$, put $T = \{4, 5\}$; then $v(T) = 2 < v(T \cup \{j\}) = 3$ for every $j \geq 6$. Since every remaining singleton has value zero, these strict interactions attach every remaining player to the same component. □

Proposition 15 (two-poorer internal-sacrifice family)**.** *For every $m \geq 2$ and $h \geq 4$ there is an integral indecomposable exact game with a poorer block P of size m, a richer suffix H of size h, and*

$$\mathrm{Lmin}(v) = D\mathbf{1}_P + 2D\mathbf{1}_H, \qquad v(H) = 2D(h - 1), \qquad D = (h - 2)(h - 3).$$

Proof. Put $r = h - 2$, write $P = \{p_1, p_2\} \dot\cup Q$ and $H = \{a, b\} \dot\cup K$, $|K| = r$. Use the common-total rows

$$\begin{aligned}
u &= De_{p_2} + D\mathbf{1}_Q + 3D(e_a + e_b) + (2r - 1)(r - 1)\mathbf{1}_K, \\
z &= 4De_{p_1} + D\mathbf{1}_Q + 2(r^2 - 1)\mathbf{1}_K, \\
w^k &= De_{p_1} + D\mathbf{1}_Q + 3D(e_a + e_b) + r(2r - 1)\mathbf{1}_{K\setminus\{k\}} \quad (k \in K), \\
\phi &= D\mathbf{1}_P + 2D\mathbf{1}_H, \\
q^j &= \phi + D(e_{p_1} - e_j) \quad (j \in Q).
\end{aligned}$$

Lemma 13 gives a normalized integral exact game. Direct substitution yields

$$\begin{aligned}
v(N \setminus \{p_2\}) &= D(m + 2h) - D, \\
v(H \setminus \{a\}) = v(H \setminus \{b\}) &= 2D(h - 1), \\
v(\{p_1, a, b\} \cup (K \setminus \{k\})) &= D(2h - 1), \\
v(H) &= 2D(h - 1).
\end{aligned}$$

At a core point x above the floor D, the complement face first fixes $x_{p_2} = D$. Write $y = x - D\mathbf{1}_N$, $E = y(Q)$, and $t = y(K)$. Then $y \geq 0$, $y_{p_2} = 0$, and $y(N) = hD$. The two co-singleton faces give

$$y_a + t \geq (h - 1)D, \qquad y_b + t \geq (h - 1)D. \tag{9}$$

The $r = h - 2$ remaining faces give, for every $k \in K$,

$$y_{p_1} + y_a + y_b + t - y_k \geq (h - 1)D.$$

Summing them and substituting $y_{p_1} + E + y_a + y_b + t = hD$ yields $t + rE \leq rD$. On the other hand, (9) and efficiency imply

$$y_{p_1} + E \leq t - rD \leq -rE.$$

Nonnegativity forces $y_{p_1} = E = 0$. Hence all of P is pinned at level D; efficiency leaves total $2Dh$ on H, equalized by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and its richer suffix has slack $2D$. The same construction is indecomposable. Indeed, p_1 has positive pair worth with every other player:

$$v(\{p_1, p_2\}) = v(\{p_1, j\}) = v(\{p_1, k\}) = D \quad (j \in Q, k \in K), \qquad v(\{p_1, a\}) = v(\{p_1, b\}) = 3D.$$

Thus the positive-pair graph is connected. □

Proposition 16 (three-poorer escape family)**.** *For every $m \geq 3$ and $h \geq 2$ there is an integral indecomposable exact game with m cumulative poorer players and richer suffix H, $|H| = h$, such that*

$$\mathrm{Lmin}(v) = \mathbf{1}_{\{p_1,p_2\}\cup Q} + 2e_{p_3} + 3\mathbf{1}_H, \qquad v(H) = 3h - 1, \qquad |Q| = m - 3.$$

Proof. Write $H = \{r_1, \ldots, r_h\}$ and take the lower envelope of

$$\begin{aligned}
a &= 2e_{p_1} + 3e_{p_3} + \mathbf{1}_Q + 5e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1,r_h\}},\\
b &= 2e_{p_2} + e_{p_3} + \mathbf{1}_Q + 4e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1\}},\\
c &= \mathbf{1}_Q + (3h+4)e_{r_h},\\
\phi &= e_{p_1} + e_{p_2} + 2e_{p_3} + \mathbf{1}_Q + 3\mathbf{1}_H,\\
q^j &= \phi + e_{p_1} - e_j \quad (j \in Q).
\end{aligned}$$

All rows have total $m + 3h + 1$, and every coordinate vanishes in a row, so Lemma 13 applies. The attained values

$$\begin{aligned}
v(N \setminus \{j\}) &= m + 3h \quad (j \in Q),\\
v(H \cup \{p_3\}) &= 3h + 2,\\
v(H \cup \{p_1\}) &= 3h + 1,\\
v(H) &= 3h - 1
\end{aligned}$$

give the progressive proof. On the face where every player receives at least one, the complement faces fix Q, the second face fixes p_1, p_2 at one, and the third fixes p_3 at two. Efficiency leaves total $3h$ on H, equalized at three by ϕ. Thus $\phi = \mathrm{Lmin}(v)$ and $s_\phi(H) = 1$. When $Q \neq \emptyset$, one member of Q has positive pair worth with every other player. When $Q = \emptyset$, the positive-pair graph connects all but p_2, and

$$v(\{p_1, p_3, r_h\}) = 4 < 5 = v(\{p_1, p_2, p_3, r_h\})$$

attaches p_2. Hence the game is indecomposable. □

Proof of Theorem 1. Corollary 7 excludes $m = 1$ for every h. Proposition 9 excludes $m \leq 3$ when $h = 1$, and Proposition 14 realizes every $m \geq 4$. Proposition 12 excludes $h \leq 3$ when $m = 2$, and Proposition 15 realizes every $m \geq 2, h \geq 4$. All remaining pairs have $m \geq 3, h \geq 2$ and are realized by Proposition 16. □

5 The four/five threshold and its mechanism

Corollary 17 (sharp player threshold)**.** *Every singleton-normalized exact game on at most four players is exact partition at* Lmin*. Five players suffice for failure.*

Proof. Every pair (m, h) with $m + h \leq 4$ is forbidden by (1). The pair $(m, h) = (3, 2)$ is realized by Proposition 16 on five players. Alternatively, $(4, 1)$ is realized by Proposition 14. □

The theorem gives two distinct first mechanisms. The $(4, 1)$ obstruction is an exterior sacrifice: the fourth poorer player supplies the first positive singleton budget capable of financing a slack richest singleton. The $(3, 2)$ obstruction is a mixed block-star mechanism. Two-poorer failures do not yet exist on five players: their active balanced packet must contain a proper rank-three trace, which first fits at $(2, 4)$ on six players. Thus "five" is not an unexplained output of enumeration; it is the first total size at which either of the two admissible boundary regions in (1) is entered.

6 What the theorem does and does not establish

The normal-cover and one-packet theorems are dimension-free: every failure is reduced to a single minimal balanced rich-trace packet and its poorer-side completion budget. The phase theorem then classifies every possible pair of boundary-side cardinalities. It does not classify all minimal balanced packets or all extreme rays of the associated two-sided coupling cone. Distinct irreducible packet classes already occur: the co-singleton class $J - I$ and a blown-up three-cycle class are not related by a single universal minor. A complete catalogue would therefore be multi-class, not a theorem that every failure contains one fixed finite configuration.

One calibration is owed to the reader. The linear-size bound in Corollary 6(i) is not itself new: it is the standard fact that a basic solution of a system of equalities has support at most the number of rows, applied to (2), whose rows are indexed by H. Sparse supports of this kind are routine in the balancedness literature surrounding the Kohlberg criterion. What is claimed here is not the bound but which single system the Lmin geometry selects, and on which prescribed suffix it lives.

The theorem's novelty remains conditional on specialist review. Exact partition, its Lorenz consequence, property I, semi-balanced inequalities, the semi-balanced/minimal-balanced correspondence, minimal balanced collections, and lower-envelope exactification are not claimed. The candidate new content is the simultaneous Lmin normal cover, the richer-side one-packet selection with its linear-size witness and slack-propagation corollary, the sharp staircase (1), and the three dimension-free indecomposable realizing families.

Reproducibility note

The proof above is human and enumerates neither games nor payoff-stratum patterns. Exact rational programs in the companion repository independently reconstruct the smallest family members, all coalition values, and staged Lmin solutions, and classify the packet systems of Remark 5 against the published catalogue of semi-balanced types. They are bug audits, not proof evidence.

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